

Lab 6

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Change 0x27 to 0x3F if LCD does not work
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0); // Column 0, Row 0  lcd.print("Hello");
}

void loop() {
  // Nothing needed here
}
```

In-Lab Question

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#define TRIG D5
#define ECHO D6
#define LED D7
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  pinMode(LED, OUTPUT);
  lcd.init();
  lcd.backlight();
  Serial.begin(9600);
}

void loop() {
  long duration;
  float distance;
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);

  duration = pulseIn(ECHO, HIGH);
  distance = duration * 0.034 / 2;

  lcd.setCursor(0, 0);
  lcd.print("Distance: ");
  lcd.print(distance);

  if (distance < 20) {
    digitalWrite(LED, HIGH);
    lcd.setCursor(0, 1);
    lcd.print("LED: ON ");
  }

  else {digitalWrite(LED, LOW);
    lcd.setCursor(0, 1);
    lcd.print("LED: OFF ");
  }  delay(500); }
```

